

Project Title: AIU Coursera Courses

Phase 1:

Application Description:

The project involves creating an extensive database system to oversee AIU Coursera Courses, covering course particulars, instructor preferences, student enrollment, and progress monitoring. This system aids in monitoring AIU courses across diverse domains, instructors' choices of Coursera courses, and student involvement. It empowers administrators to handle program invitations, instructors to align Coursera modules with course weeks, and students to register and finish courses. Vital features comprise generating semester reports on course requests, student enrollments, and progress, weekly updates on license distribution and student withdrawals, and end-of-semester/year reports on student accomplishments. The database will represent AIU course specifics, instructor preferences, Coursera course details, student enrollment information, and completion data, ensuring thorough tracking and reporting capabilities for stakeholders with different access levels.

Business Rules:

1. Each AIU course must have at least one main instructor designated for the semester.
2. The main instructor for an AIU course must have administrative access to the associated Coursera program for that course.
3. The duration of a Coursera course selected for an AIU course must align with the duration of the corresponding semester.
4. Students can only be added to the Coursera program for an AIU course if they are registered for that AIU course.
5. Coursera courses used in previous semesters for an AIU course should be easily accessible to the main instructor for reference and reuse.
6. Each AIU field must have its own designated Coursera program, administered by the respective field coordinator.

7. Students who fail to enroll in any Coursera course within a specified timeframe should be remove from the AIU course.
8. The system should provide automated reminders to instructors regarding upcoming deadlines for inviting students and managing course enrollments.
9. Coursera course completion data should be periodically updated in the database to track student progress and assess the effectiveness of the AIU-Coursera integration.
10. Field coordinators should have the authority to approve Coursera course selections and mappings proposed by instructors within their respective fields.

Potential Queries:

1. Which AIU courses lack a designated main instructor this semester?
2. Are there main instructors without administrative access to their AIU course's Coursera program?
3. Do any Coursera courses selected for AIU courses have durations mismatched with the semester duration?
4. Which registered students haven't been added to their AIU course's Coursera program?
5. What Coursera courses were previously utilized for specific AIU courses, accessible to main instructors?
6. Which AIU fields lack their own designated Coursera programs, and who are their coordinators?
7. Which students failed to enroll in any Coursera course within the specified timeframe for their AIU courses?
8. What are the upcoming deadlines and automated reminders needed for managing course enrollments and student invitations?

Phase 2:

Entities:

1. Coursera Courses
2. Aiu Courses
3. Instructor
4. Admin
5. Student
6. Program
7. Field
8. License

9. Enrollment
10. Module

Relations And Cardinalities:

1. **One to Many** Between Coursera Courses and Aiu Courses → **(Belong to)**
2. **Many to Many** Between Coursera Courses and Instructor → **(Work At)**
3. **Many to Many** Between Coursera Courses and Student → **(Learn)**
4. **One to Many** Between Coursera Courses and License → **(Have)**
5. **One to Many** Between Coursera Courses and Enrollment → **(Have Many)**
6. **Many to Many** Between Aiu Courses and Instructor → **(Teach Them)**
7. **Many to One** Between Aiu Courses and Field → **(Includes)**
8. **One to Many** Between Aiu Courses and Module → **(Have Much)**
9. **Many to Many** Between Aiu Courses and Student → **(Enroll)**
10. **Many to Many** Between Instructor and Student → **(Teach)**
11. **Many to One** Between Instructor and Program → **(Teach in)**
12. **One to Many** Between Student and License → **(Has)**
13. **One to Many** Between Student and Enrollment → **(Has Many)**
14. **Many to One** Between Student and Program → **(Belongs to)**
15. **One to Many** Between Field and Program → **(Has Much)**
16. **One to Many** Between Admin and Program → **(Administrate)**

Attributes:

Coursera Courses:

Coursersra Course code (Primary Key) , Coursera Course Name , Coursera Course Field , Enrollment Date , Deadline

Aiu Courses:

Aiu Course code (Primary Key) , Aiu Course Name , Aiu Field , Enrollment Date , Deadline

Instructor:

Instructor ID (Primary Key) , Instructor Name , Instructor Age , Field

Student:

Student ID (Primary Key) , Student Name , Level , Field , Adress (House No. , Street , PIN) , Student Age(Derived Attribute) , Date of Birth , Email

Admin:

Admin ID (Primary Key) , Admin Name , Program

Field:

Faculty code (Primary Key) , Faculty Name , Field Program , Disjoint (CS , Engineering , Pharmacy , Dentistry)

Program:

Program Code (Primary Key) , Program Name , No. of Students , No. of Instructors , No. of Courses

License:

License Course Code (Primary Key) , License Course Name , License Date , No. of Licenses

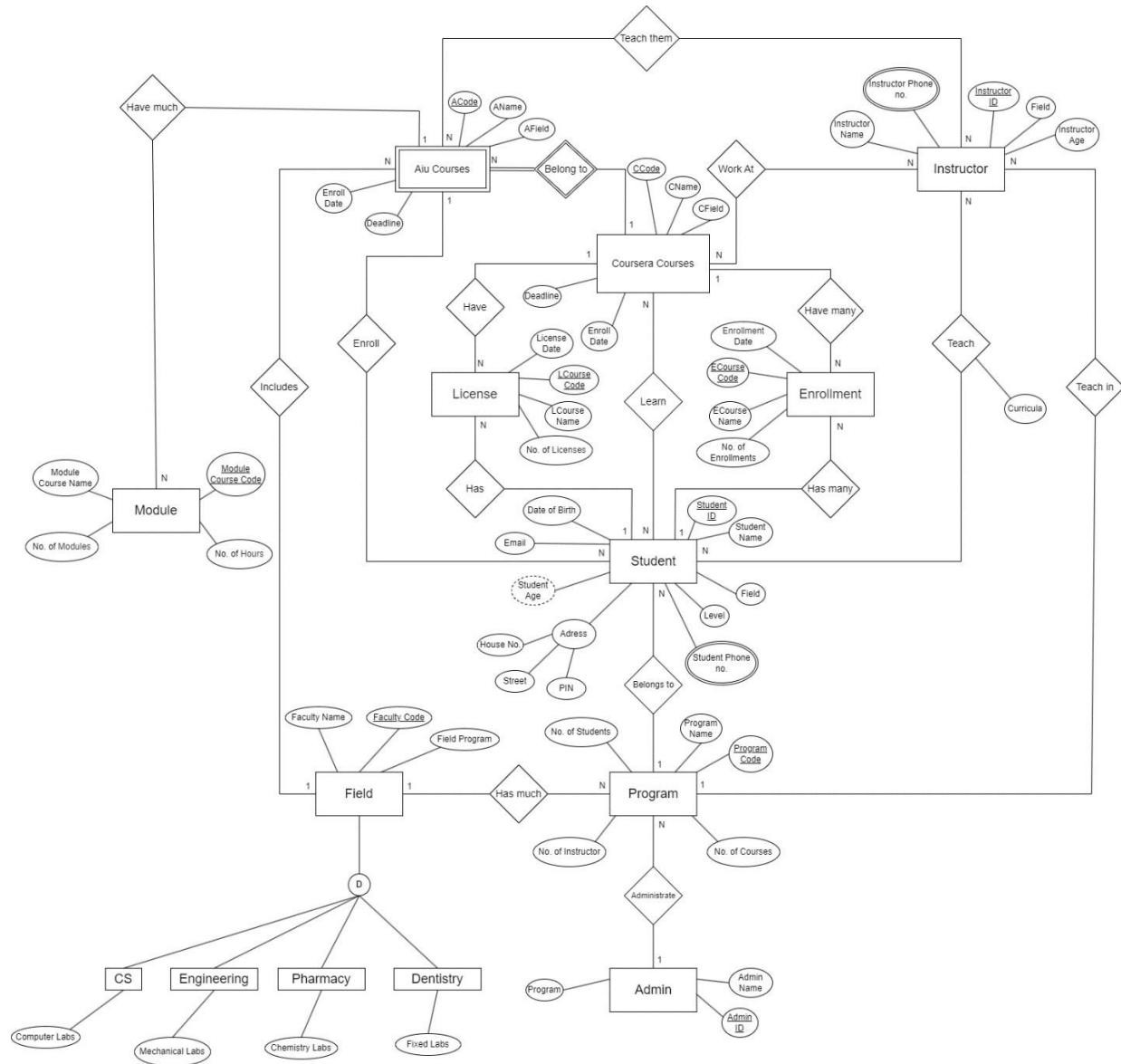
Enrollment:

Enrollment Course Code (Primary Key) , Enrollment Course Name , Enrollment Date , No. of Enrollments

Module:

Module Course Code (Primary Key) , Module Course Name , No. of Modules , No. of Hours

ERD:



Phase 3:

Schema Normalization:

ONF:

Coursera Courses (Coursersra Course code : **Int** , Coursera Course Name : **String** , Coursera Course Field : **String** , Enrollment Date : **Date** , Deadline : **Date**)

Aiu Courses (Aiu Course code : **Int** , Coursersra Course code : **Int** , Faculty code : **Int** , Aiu Course Name : **String** , Aiu Field : **String** , Enrollment Date : **Date** , Deadline : **Date**)

Instructor (Instructor ID : **Int** , Program Code : **Int** , Instructor Name : **String** , Instructor Age : **Int** , Field : **String** , Instructor Phone no. : **String**)

Student (Student ID : **Int** , Program Code : **Int** , Aiu Course code : **Int** , Student Name : **String** , Level : **Int** , Field : **String** , House No. : **Int** , Street : **String** , PIN : **Int** , Date of Birth : **Date** , Email : **String** , Student Phone no. : **String**)

Admin (Admin ID : **Int** , Admin Name : **String** , Program : **String**)

Field (Faculty code : **Int** , Faculty Name : **String** , Field Program : **String**)

CS (Faculty code : **Int** , Computer Labs : **String**)

Engineering (Faculty code : **Int** , Mechanical Labs : **String**)

Pharmacy (Faculty code : **Int** , Chemistry Labs : **String**)

Dentistry (Faculty code : **Int** , Fixed Labs : **String**)

Program (Program Code : **Int** , Admin ID : **Int** , Program Name : **String** , No. of Students : **Int** , No. of Instructors : **Int** , No. of Courses : **Int**)

License (License Course Code : **Int** , Student ID : **Int** , Coursersra Course code : **Int** , License Course Name : **String** , License Date : **Date** , No. of Licenses : **Int**)

Enrollment (Enrollment Course Code : **Int** , Student ID : **Int** , Coursersra Course code : **Int** , Enrollment Course Name : **String** , Enrollment Date : **Date** , No. of Enrollments : **Int**)

Module (Module Course Code : **Int** , Aiu Course code : **Int** , Module Course Name : **String** , No. of Modules : **Int** , No. of Hours : **Int**)

Work At (Coursersra Course code : **Int** , Instructor ID : **Int**)

Learn (Coursersra Course code : **Int** , Student ID : **Int**)

Teach Them (Aiu Course code : **Int** , Instructor ID : **Int**)

Enroll (Aiu Course code : **Int** , Student ID : **Int**)

Teach (Instructor ID : **Int** , Student ID : **Int** , Curricula : **String**)

1NF:

Coursera Courses (Cousersra Course code , Coursera Course Name , Coursera Course Field , Enrollment Date , Deadline)

Aiu Courses (Aiu Course code , Cousersra Course code , Faculty code , Aiu Course Name , Aiu Field , Enrollment Date , Deadline)

Instructor (Instructor ID , Program Code , Instructor Name , Instructor Age , Field)

Student (Student ID , Program Code , Aiu Course code , Student Name , Level , Field , House No. , Street , PIN , Date of Birth , Email)

Admin (Admin ID , Admin Name , Program)

Field (Faculty code , Faculty Name , Field Program)

CS (Faculty code , Computer Labs)

Engineering (Faculty code , Mechanical Labs)

Pharmacy (Faculty code , Chemistry Labs)

Dentistry (Faculty code , Fixed Labs)

Program (Program Code , Admin ID , Program Name , No. of Students , No. of Instructors , No. of Courses)

License (License Course Code , Student ID , Cousersra Course code , License Course Name , License Date , No. of Licenses)

Enrollment (Enrollment Course Code , Student ID , Cousersra Course code , Enrollment Course Name , Enrollment Date , No. of Enrollments)

Module (Module Course Code , Aiu Course code , Module Course Name , No. of Modules , No. of Hours)

Work At (Cousersra Course code , Instructor ID)

Learn (Cousersra Course code , Student ID)

Teach Them (Aiu Course code , Instructor ID)

Enroll (Aiu Course code , Student ID)

Teach (Instructor ID , Student ID , Curricula)

Contact Student (Student ID , Student Phone no.)

Contact Instructor (Instructor ID , Instructor Phone no.)

2NF:

Coursera Courses (Cousersra Course code , Coursera Course Name , Coursera Course Field , Enrollment Date , Deadline)

Aiu Courses (Aiu Course code , Cousersra Course code , Faculty code , Aiu Course Name , Aiu Field , Enrollment Date , Deadline)

Instructor (Instructor ID , Program Code , Instructor Name , Instructor Age , Field)

Student (Student ID , Program Code , Aiu Course code , Student Name , Level , Field , Date of Birth , Email , House No. , Street , PIN)

Admin (Admin ID , Admin Name , Program)

Field (Faculty code , Faculty Name , Field Program)

CS (Faculty code , Computer Labs)

Engineering (Faculty code , Mechanical Labs)

Pharmacy (Faculty code , Chemistry Labs)

Dentistry (Faculty code , Fixed Labs)

Program (Program Code , Admin ID , Program Name , No. of Students , No. of Instructors , No. of Courses)

License (License Course Code , Student ID , Cousersra Course code , License Course Name , License Date , No. of Licenses)

Enrollment (Enrollment Course Code , Student ID , Cousersra Course code , Enrollment Course Name , Enrollment Date , No. of Enrollments)

Module (Module Course Code , Aiu Course code , Module Course Name , No. of Modules , No. of Hours)

Work At (Cousersra Course code , Instructor ID)

Learn (Coursersra Course code , Student ID)

Teach Them (Aiu Course code , Instructor ID)

Enroll (Aiu Course code , Student ID)

Teach (Instructor ID , Student ID , Curricula)

Contact Student (Student ID , Student Phone no.)

Contact Instructor (Instructor ID , Instructor Phone no.)

THE SCHEMA IS ALREADY 2NF NORMALIZED

3NF:

Coursera Courses (Coursersra Course code , Coursera Course Name , Coursera Course Field , Enrollment Date , Deadline)

Aiu Courses (Aiu Course code , Coursersra Course code , Faculty code , Aiu Course Name , Aiu Field , Enrollment Date , Deadline)

Instructor (Instructor ID , Program Code , Instructor Name , Instructor Age , Field)

Student (Student ID , Program Code , Aiu Course code , Student Name , Level , Field)

Admin (Admin ID , Admin Name , Program)

Field (Faculty code , Faculty Name , Field Program)

CS (Faculty code , Computer Labs)

Engineering (Faculty code , Mechanical Labs)

Pharmacy (Faculty code , Chemistry Labs)

Dentistry (Faculty code , Fixed Labs)

Program (Program Code , Admin ID , Program Name)

License (License Course Code , Student ID , Coursersra Course code , License Course Name , License Date)

Enrollment (Enrollment Course Code , Student ID , Coursersra Course code , Enrollment Course Name , Enrollment Date)

Module (Module Course Code , Aiu Course code , Module Course Name)

Work At (Cousersra Course code , Instructor ID)

Learn (Cousersra Course code , Student ID)

Teach Them (Aiu Course code , Instructor ID)

Enroll (Aiu Course code , Student ID)

Teach (Instructor ID , Student ID , Curricula)

Contact Student (Student ID , Student Phone no.)

Contact Instructor (Instructor ID , Instructor Phone no.)

Student Info (Student Name , Date of Birth , Email , House No. , Street , PIN)

Course's Module (Module Course Name , No. of Modules , No. of Hours)

Course's Enrollments (Enrollment Course Name , No. of Enrollments)

Course's Licenses (License Course Name , No. of Licenses)

Data of Program (Program Name , No. of Students , No. of Instructors , No. of Courses)

Phase 4:

Business questions:

1. Which courses have the highest enrollment?
2. What is the average number of modules in each AIU course?
3. Which instructor teaches the most courses?
4. What is the distribution of students across different fields?
5. Which program has the highest number of instructors?
6. What is the average age of instructors in each field?

Relational Algebra:

1. $\pi_{(\text{CourseraCourseName}, \text{EnrollmentCount})}(\gamma_{(\text{CourseraCourseName}, \text{COUNT}(\text{StudentID}))}(\sigma_{(\text{CourseraCourses.CourseraCourseCodeEnrollment.CourseraCourseCode})}(\text{CourseraCourses} \bowtie \text{Enrollment}))$
2. $\pi_{(\text{AiuCourseName}, \text{AvgModules})}(\gamma_{(\text{AiuCourseName}, \text{AVG}(\text{NoOfModules}))}(\sigma_{(\text{AiuCourses.AiuCourseCodeModule.AiuCourseCode})}(\text{AiuCourses} \bowtie \text{Module}))$
3. $\pi_{(\text{InstructorName}, \text{CoursesTaught})}(\gamma_{(\text{InstructorName}, \text{COUNT}(\text{InstructorID}))}(\sigma_{(\text{Instructor.InstructorID} = \text{TeachThem.InstructorID})}(\text{Instructor} \bowtie \text{TeachThem}))$
4. $\pi_{(\text{Field}, \text{StudentCount})}(\gamma_{(\text{Field}, \text{COUNT}(\text{StudentID}))}(\text{Student}))$
5. $\pi_{(\text{ProgramName}, \text{NoOfInstructors})}(\text{Program})$
6. $\pi_{(\text{Field}, \text{AvgAge})}(\gamma_{(\text{Field}, \text{AVG}(\text{InstructorAge}))}(\text{Instructor}))$